

In Lab 01 Tasks

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Assume yourself as a **Server Administrator**. Several users report that they cannot access the company's file server.

The file server IP address is **172.16.5.43**.

Your first task is to verify that the server is accessible from a user's computer.

Task # 01

Which command will you use to test connectivity with the file server? Write the complete command.

> ping 172.16.5.43

```
C:\Users\k240512>ping 172.16.5.43

Pinging 172.16.5.43 with 32 bytes of data:
Reply from 172.16.5.43: bytes=32 time<1ms TTL=127
Reply from 172.16.5.43: bytes=32 time=1ms TTL=127
Reply from 172.16.5.43: bytes=32 time<1ms TTL=127
Reply from 172.16.5.43: bytes=32 time=1ms TTL=127

Ping statistics for 172.16.5.43:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Investigating an issue with the company's website:

www.flex.nu.edu.pk

The website is not opening using its domain name.

Task # 02

Which command would you use to find the IP address associated with www.flex.nu.edu.pk?

> nslookup flex.nu.edu.pk

user can access the Internet, but accessing a remote company server is extremely slow.

You want to determine **which routers/hops the packets are passing through** before reaching the destination.

The server is 8.8.8.8.

Task # 03

Which command should you use on Windows?

> tracert 8.8.8.8

```
C:\Users\k240512>tracert 8.8.8.8
```

```
Tracing route to dns.google [8.8.8.8]  
over a maximum of 30 hops:
```

```
  1      1 ms      1 ms      6 ms  172.16.205.1  
  2     <1 ms     <1 ms     <1 ms  172.16.5.5  
  3      1 ms     <1 ms      1 ms  GPONUser38229-1.wateen.net [110.38.229.1  
] 4      1 ms      1 ms      1 ms  10.17.18.114  
  5      1 ms      1 ms      1 ms  10.17.18.102  
  6      1 ms      1 ms      1 ms  10.17.12.7  
  7      1 ms      1 ms      1 ms  10.17.12.4  
  8      2 ms      2 ms      2 ms  119.63.130.114  
  9      2 ms      2 ms      2 ms  110.93.252.172  
 10      2 ms      3 ms      2 ms  110.93.252.222  
 11     16 ms     17 ms     18 ms  142.250.165.127  
 12     18 ms     17 ms     17 ms  216.239.41.109  
 13     18 ms     17 ms     18 ms  172.253.79.255  
 14     16 ms     16 ms     16 ms  dns.google [8.8.8.8]
```

```
Trace complete.
```

IP Address: 127.0.0.1 What is this IP used for? > **loopback address**

The **Data Link layer** uses **MAC addresses** and devices like **Switches**.

The **Network layer** is responsible for **logical addressing** and uses devices like **routers**.